Predicting MBTI Personality Types from Text Using Retrieval-Based and Classification-Based Approaches

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Abstract

We frame MBTI personality prediction as a *text retrieval and ranking* problem: given a user's writing as a query, the system retrieves the best-matching type from 16 candidates. We compare three IR-grounded approaches: (1) BM25 [7] ranking against per-type document profiles, (2) TF-IDF [6] features with Logistic Regression and Naive Bayes, and (3) dense sentence-embedding cosine similarity. [9] [8] We further decompose the problem into four per-dichotomy binary classifiers (I/E, S/N, T/F, J/P) to reveal which dimensions are most distinguishable from text, and deploy the pipeline as a personality-aware AI assistant. Evaluation uses the Kaggle MBTI dataset [2] with macro F1 [5] and dichotomy confusion analysis.

1 Introduction

The Myers-Briggs Type Indicator (MBTI) [4] classifies individuals along four binary dimensions—Introversion/Extraversion (I/E), Sensing/Intuition (S/N), Thinking/Feeling (T/F), and Judging/Perceiving (J/P)—yielding 16 personality types. Since language correlates with personality, each type can be viewed as a "document collection" with distinctive vocabulary, and unseen text as a *query* matched via IR similarity—directly applying vector-space models, BM25, and dense retrieval. The problem's complexity arises from (a) fine-grained 16-class distinctions with overlapping signals, (b) noisy social-media text, and (c) class imbalance. We also perform per-dichotomy analysis to identify which dimensions are hardest to distinguish.

2 Summary of Proposed Work

Software. We build a Python system with three IR-grounded approaches. **(1) BM25 Retrieval:** [7] We concatenate all training posts per MBTI type into 16 type-level documents and rank them against query text via BM25 (`rank_bm25`)—directly applying document retrieval. **(2) TF-IDF + Classification:** TF-IDF [6]features (unigrams + bigrams, sublinear TF) feed Logistic Regression and Multinomial Naive Bayes (scikit-learn). **(3) Dense Retrieval:** [3] Posts are encoded with a sentence-transformer (`all-MiniLM-L6-v2`) [1]; per-type centroid embeddings classify via cosine similarity. We also train Logistic Regression on embedding features.

We implement a **per-dichotomy binary decomposition**: four independent binary classifiers each separating one MBTI dimension (e.g., all I-types vs. E-types) on TF-IDF [6] features; the four outputs are concatenated into a 4-letter type. This reveals per-dimension accuracy and top discriminative features per axis. A Streamlit front-end serves as a personality-aware AI assistant: users enter text, the system predicts their MBTI type, then tailors responses and recommends topics aligned with the inferred personality.

Data. The *MBTI Personality Type Dataset* from Kaggle [2] (~8,600 users, 50 posts each, self-reported labels). Preprocessing: URL removal, lowercasing, stopword removal, lemmatization. Stratified 80/20 train-test split.

Evaluation. Macro F1 [5] and per-class precision/recall. A *dichotomy confusion analysis* records which dimensions flipped per error (e.g., INFP→ENFP flips I/E); combined with per-dichotomy binary accuracy, this reveals where and why errors occur.

Timeline and Division of Work (5 members, ~30 hours total):

Who	Task	Hrs	Due
Xinyan Chen	Data collection, cleaning, EDA + report	6	Week 6
Kaiyu Liu	BM25 retrieval + TF-IDF classifiers (LR/NB)	6	Week 8
Zhicheng Liao	Embedding pipeline (cosine + LR)	6	Week 8
Yuyang Zeng	Dichotomy binary classifiers + confusion analysis	6	Week 10
Mianfeng Liu	Front-end demo + presentation	6	Week 11

All members participate in milestone reporting and peer review. Integration testing occurs in Weeks 11–12.

References

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